

TWO-MONTH LIVE PROGRAMME · AGES 13-18

From Zero to *AI Builder*

In two months, go from your very first line of code to building and shipping real, AI-powered apps — websites, JavaScript apps, Python, machine learning, and apps powered by large language models.

32 classes 8 weeks · 4 / week Live online · mentor-led Project-based Certificate

THE PROGRAMME

What this is

A fast, hands-on two-month programme where teens go from **knowing nothing about code to building and launching their own AI-powered apps**. Every single class builds something real — no endless theory.

We start on the web (HTML, CSS and **JavaScript**) so students see results on screen from day one. Then we move into **Python**, the language of AI. Finally we go deep into the exciting part: **machine learning and building apps with LLMs** — chatbots, AI tools, "chat with your own notes", and a full-stack AI product — connecting everything using API keys and real AI models.

Three phases, one journey. Phase 1 · Web & JavaScript (Weeks 1–3) → build and deploy interactive apps · **Phase 2 · Python** (Weeks 4–5) → real programming and automation · **Phase 3 · AI, ML & LLM apps** (Weeks 6–8) → build and ship your own AI product.

WHO IT'S FOR

Made for teenage builders (13–18)

- **Total beginners** — no coding experience needed at all
- **Curious teens** who love tech, games, or AI and want to build, not just use
- **Students** who want a real portfolio of projects before college
- **Future creators & founders** with app or startup ideas
- **Scratch / Python graduates** ready to level up to real apps and AI
- **Anyone 13–18** who wants to actually ship something cool

IN PRACTICE

What you'll build

By the end you'll have a portfolio of real projects you can show off and share with a link.

Highlights:

WEBSITES & LANDING PAGES

- A personal "about me" site and a polished landing page

LIVE API-POWERED APPS

- A weather / movie app that pulls real data, deployed to a URL

MACHINE-LEARNING APPS

- A model that predicts something from real data

AI WITH YOUR OWN DATA

- A "chat with my notes / PDF" app and an AI agent that uses tools

INTERACTIVE WEB APPS & GAMES

- A quiz app, a to-do list and a number-guessing game

PYTHON AUTOMATIONS

- A web-scraper or auto-report script that does the boring work

AI CHATBOTS & LLM TOOLS

- Your own chatbot with a personality and an AI study-helper

CAPSTONE AI PRODUCT

- A full-stack AI app, deployed live, that you present and share

COVERAGE

Tools & skills you'll learn

- **Web** — HTML, CSS, responsive design
- **JavaScript** — logic, the DOM, interactive apps
- **Git & GitHub** + deploying live to the web
- **Python** — the language of AI, from scratch
- **APIs & JSON** — connecting apps to the internet
- **Data** — pandas, charts and real datasets
- **Machine learning** — train and use your own model
- **AI & LLMs** — API keys, prompting, RAG, agents, images/voice

Week-by-week syllabus

Four live classes a week for eight weeks. Each class lists what you learn and what you build.

PHASE 1 • WEEKS 1-3

Web & JavaScript — see results from day one

Build and deploy interactive web apps while learning how code really works.

WEEK 1 Your first web pages — HTML & CSS

01 How the Web Works + Your First Page

YOU'LL LEARN

What happens when you open a website; HTML basics — headings, text, links and images; using a code editor.

YOU'LL BUILD

Your first live web page — an "about me".

02 Structure & Forms

YOU'LL LEARN

Page structure, lists, buttons, forms and inputs, and semantic HTML.

YOU'LL BUILD

A profile page with a working contact form.

03 Styling with CSS

YOU'LL LEARN

Colors, fonts, spacing and the box model; classes and selectors.

YOU'LL BUILD

Restyle your page so it actually looks great.

04 Layouts That Look Pro

YOU'LL LEARN

Flexbox and grid; responsive design that works on phone and laptop.

YOU'LL BUILD

A clean landing page for an idea you love.

WEEK 2 JavaScript — make it interactive

05 JavaScript Basics

YOU'LL LEARN

Variables, text and numbers, operators; running JavaScript in the browser.

YOU'LL BUILD

A tip / score calculator.

06 Logic & Functions

YOU'LL LEARN

`if / else`, loops, and writing your own reusable functions.

YOU'LL BUILD

A number-guessing game.

07 The DOM — Bring Pages to Life

YOU'LL LEARN

Selecting elements, handling clicks and events, and changing the page live.

YOU'LL BUILD

An interactive to-do list.

08 Project — Interactive Quiz App

YOU'LL LEARN

Putting it together — app state, events and updating the UI.

YOU'LL BUILD

A multi-question quiz app that scores you.

WEEK 3 Real apps — data, APIs & going live

09 Working with Data — Arrays & Objects

YOU'LL LEARN

Storing lists and structured data; looping over data to build a UI automatically.

YOU'LL BUILD

A card gallery generated from data.

10 Talking to the Internet — APIs & JSON

YOU'LL LEARN

What an API is; `fetch`; reading JSON responses.

YOU'LL BUILD

An app that pulls live data from a public API.

11 Async JavaScript

YOU'LL LEARN

Promises, `async / await`, loading states and handling errors.

YOU'LL BUILD

A search app with live results.

12 Ship It — Git, GitHub & Deploy

YOU'LL LEARN

Saving your work with Git & GitHub, then deploying to the web.

YOU'LL BUILD

Your app live on a real URL you can share.

Python — the language of AI

Learn real programming and automation, and get ready for machine learning.

WEEK 4 Python foundations

13 Meet Python

YOU'LL LEARN

Why Python; setup; variables, types, and input / output.

YOU'LL BUILD

An interactive text-based mini-app.

14 Logic in Python

YOU'LL LEARN

Conditionals, loops, and writing clean, readable code.

YOU'LL BUILD

A quiz or calculator in Python.

15 Functions & Modules

YOU'LL LEARN

Reusable functions, importing modules, and organizing your code.

YOU'LL BUILD

A small toolkit of your own functions.

16 Data Structures

YOU'LL LEARN

Lists, dictionaries and sets — and when to use each.

YOU'LL BUILD

A command-line contact book or inventory.

WEEK 5 Python for real work

17 Files & Real Data (JSON)

YOU'LL LEARN

Reading and writing files; working with JSON data that persists.

YOU'LL BUILD

A note-saver or expense tracker that remembers.

18 Libraries & Environments

YOU'LL LEARN

`pip`, installing packages, and virtual environments.

YOU'LL BUILD

Something fun with a library — a QR code or an edited image.

19 Python Meets APIs

YOU'LL LEARN

The `requests` library; calling APIs and handling responses in Python.

YOU'LL BUILD

A script that fetches and shows live data.

20 Project — Python Automation

YOU'LL LEARN

Automating a boring task end-to-end.

YOU'LL BUILD

A mini web-scraper or an auto-report generator.

PHASE 3 • WEEKS 6-8

AI, Machine Learning & LLM Apps — the exciting part

Understand how AI works, then build and ship your own AI-powered product.

WEEK 6 Into AI & machine learning

21 What AI & ML Really Are

YOU'LL LEARN

AI vs machine learning vs deep learning; how machines "learn" from data — with real examples, no hype.

YOU'LL BUILD

Explore a real dataset and spot patterns.

22 Data Superpowers — pandas

YOU'LL LEARN

Loading, cleaning and exploring data with pandas; making simple charts.

YOU'LL BUILD

Analyze a real dataset and chart your findings.

23 Your First ML Model

YOU'LL LEARN

Features & labels, train / test split, and making predictions with scikit-learn.

YOU'LL BUILD

A model that predicts something real.

24 Project — Prediction App

YOU'LL LEARN

Checking if your model is any good, then using it on brand-new input.

YOU'LL BUILD

A small app that makes predictions from your model.

WEEK 7 Building with LLMs — the fun part

25 How LLMs Work + Your First API Key

YOU'LL LEARN

What large language models do; getting and safely protecting an API key.

YOU'LL BUILD

Your very first call to an AI model from code.

26 Build an AI Chatbot

YOU'LL LEARN

Sending prompts, reading responses, and building a chat loop.

YOU'LL BUILD

Your own chatbot with its own personality.

27 Prompt Engineering & Structured Output

YOU'LL LEARN

Writing great prompts and getting clean JSON back to power a real app.

YOU'LL BUILD

An AI tool — a study-helper or idea generator.

28 AI Superpowers — Images & Voice

YOU'LL LEARN

Sending images to AI, generating images, and text-to-speech basics (multimodal AI).

YOU'LL BUILD

An app that describes or creates images.

WEEK 8 Connect everything & ship your AI product

29 Chat With Your Own Data (RAG)

YOU'LL LEARN

Embeddings and retrieval — letting AI answer questions from your own documents.

YOU'LL BUILD

A "chat with my notes / PDF" app.

30 AI Agents — Apps That Take Action

YOU'LL LEARN

Connecting tools and APIs so AI can do things, not just talk.

YOU'LL BUILD

A mini AI agent that uses a real tool.

31 Full-Stack AI App

YOU'LL LEARN

Connecting a web front-end to a Python + LLM backend into one app.

YOU'LL BUILD

A polished AI web app, end-to-end.

32 Capstone — Ship Your AI Product + Showcase

YOU'LL LEARN

Planning, building, testing and deploying a real product you're proud of.

YOU'LL BUILD

Your own deployed AI app with a shareable URL, which you present — and earn your Modern Age Coders certificate.

OUTCOMES

What you'll be able to do

- Build and **deploy your own websites and web apps** to a real URL.
- Write real programs in **JavaScript and Python**.
- Pull **live data from APIs** and build apps around it.
- Train and use a **simple machine-learning model**.
- Build **AI apps with LLMs** — chatbots, tools, "chat with your data", and agents.
- Ship a **capstone AI product** with a shareable link for your portfolio.
- Earn the Modern Age Coders **certificate** of completion.

REQUIREMENTS

What you need to start

DEVICE	Any laptop or desktop (Windows, Mac or Linux) with internet — nothing powerful needed.
EXPERIENCE	None at all — we start from your very first line of code.
ACCOUNTS	A free GitHub account, and free / low-cost AI access for the LLM classes — we guide students and parents through safe setup.
ATTITUDE	Curiosity and a willingness to build. That's it.

FORMAT & NEXT STEP

Delivery & logistics

FORMAT	Two-month programme — 32 live classes, 4 per week over 8 weeks.
DELIVERY	Live online, mentor-led and project-based; every class builds something real.
SUPPORT	Small batches, doubt-clearing, and class recordings so nothing is missed.
CERTIFICATION	Modern Age Coders certificate on completion, plus a portfolio of projects.

Enrol your teen — or book a free demo class

Pricing available on request. Group and 1-on-1 options; flexible batches for school schedules.

Modern Age Coders

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